

Muhammad Bilal

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TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C, PostgreSQL, MySQL, HTML, CSS

Frameworks: React, Node.js, Spring Boot, Angular, Bootstrap, Next.js, Express.js

Developer Tools: Git, Docker, Postman, VSCode, IntelliJ, Maven, Kubernetes, Insomnia, Power BI, Linux, Jira, Agile, Scrum

Libraries: Pandas, NumPy, Matplotlib, Tkinter, Scikit-learn, MyBatis

EDUCATION

University of Massachusetts Amherst

Bachelor of Science in Computer Science

Expected Graduation: May 2026

EXPERIENCE

Fidelity Investments

June 2025 – August 2025

Software Engineer Intern

Smithfield, RI

- Integrated ServiceNow Change Management functionality into an internal admin application used for Single Sign-On (SSO) configurations, ensuring over 150+ annual Production changes are accurately logged and compliant
- Built and connected the ServiceNow Configuration Management Database (CMDB) to the Spring Boot backend, enabling real-time retrieval and updates of configuration items (CIs)
- Developed a dynamic Angular-based “Edit” feature on the admin UI to enable direct modification of SSO client connections and streamline administrative workflows
- Streamlined and secured real-time SSO updates using TypeScript-driven form handling and Spring API integration, reducing manual errors by ensuring accurate logging of Production changes
- Practiced Agile methodology by actively leading and participating in Scrum ceremonies, utilizing Jira and Confluence to manage work items and document technical changes

Edworldo

June 2023 – August 2023

Software Engineer Intern

Austin, TX

- Built a prototype question generator that crafted over 1,500 practice questions mimicking STARR test formats, improving test prep effectiveness by 45% in internal trials
- Trained and fine-tuned advanced machine learning models using over 20,000 labeled educational data points to ensure question relevance and accuracy
- Implemented robust error handling and optimized RESTful API integrations to improve reliability and streamline front-end to back-end communication

Persivia Inc

June 2021 – August 2021

Machine Learning Intern

Marlborough, MA

- Developed ML models in Python to predict patient costs with 87%+ accuracy, aiding hospitals in reducing cost inefficiencies
- Used NumPy and Pandas to process datasets of 50,000+ patient records, analyzing trends by age, condition severity, and disease risk levels
- Leveraged Power BI to build interactive reports for cost prediction models, enabling easier interpretation and faster insights

PROJECTS

Chef Krrish Chatbot | *React.js, Next.js, Firebase, Tailwind CSS, OpenRouter API*

- Built a full-stack dietary chatbot web app using Firebase Auth, Firestore, and OpenRouter API, helping over 100+ users customize nutritional guidance based on allergies and goals
- Implemented chat history, dark mode, and dietary preference features, enhancing UX and increasing user retention by 30% during beta testing

A.I Personal Assistant | *Python, OpenAI, gTTS, pyaudio, playsound*

- Created a real-time voice assistant that processed voice commands and generated AI responses with 95%+ speech recognition accuracy
- Converted over 250+ user queries into audio responses using OpenAI GPT-3.5 and Google Text-to-Speech, improving accessibility and interactivity

Quiz Maker | *Python, Jupyter Notebook, OpenAI, Gradio, ipywidgets*

- Generated 500+ auto-created quiz questions across subjects using OpenAI's API, streamlining test creation for educators
- Designed an interactive UI in Jupyter with Gradio and ipywidgets, boosting usability and enabling scalable deployment to schools

RELEVANT COURSES & CERTIFICATIONS

- Courses: Data Structures and Algorithms, Software Engineering, Object-Oriented Programming, Linear Algebra, Discrete Math
- Certifications: Machine Learning Specialization (Coursera), Learn Java Course (Codecademy), Learn C Course (Codecademy)